

World models learn their simulator’s integrator, not the physics it approximates

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Abstract

A world model trained only on pendulum video learns a conserved quantity — but not the one physics prescribes. It learns the shadow Hamiltonian of the simulator’s integrator, $\tilde{H} = H + \frac{\Delta t}{2} mg \frac{l}{2} \dot{\theta} \sin \theta$, whose correction coefficient is a property of the discretisation rather than of pendulums. We test that directly by varying the simulator’s timestep: across a $4\times$ range, the coefficient the model recovers tracks the integrator’s prediction with a slope of 2.484 ± 0.058 against a parameter-free 2.500, with the minimum landing exactly on prediction for 10 of 12 models and a wrong-sign control beaten 12 of 12. As the timestep coarsens, textbook energy becomes progressively worse conserved by the model ($\Delta t^{1.41}$) while the shadow quantity stays at the model’s own floor, so the gap between them grows from $1.03\times$ to $13.5\times$. This has a sharp consequence for interpretability. A polynomial readout fitted to true physical energy reaches a correlation of 0.9999 — yet it is $6.7\times$ less preserved by the model’s own transition than a label-free scalar, and enforcing it during imagination makes the model’s physics worse, while the label-free scalar improves it by 42%. The perfect probe is fitted to a quantity the generating process does not conserve, and no amount of probe hygiene detects this, because nothing is wrong with the probe. Whether a latent encodes a quantity and whether the dynamics preserve it are different properties, and they dissociate on five axes — untrained models, out-of-distribution states, a second architecture, an untrained transition, and actuation — always with decodability surviving where dynamical structure fails. A statistic on the transition operator separates them in every case, and the quantity it selects is causally deployed: enforcing it cuts drift in decoded physical energy by 55–76% on unseen trajectories, with 0 of 60 magnitude-matched random directions beating it, and the effect survives an interchange 50 steps into autonomous imagination. We also report where it stops: the correction confers no benefit to a planner, and we measure why.

1 Introduction

A world model trained on simulated video is trained on the output of an integrator. We show it learns the integrator.

On a DreamerV3 world model trained only on pendulum video, a label-free search recovers a conserved scalar — and that scalar is not textbook energy but the shadow Hamiltonian of the simulator’s numerical scheme, $\tilde{H} = H + \frac{\Delta t}{2} mg \frac{l}{2} \dot{\theta} \sin \theta$ (Hairer et al., 2006a). The correction coefficient is a property of the discretisation, not of pendulums, so it makes a parameter-free prediction: vary the simulator’s timestep and the recovered coefficient must follow. It does. Across a $4\times$ range of Δt the recovered coefficient tracks the integrator’s prediction with slope 2.484 ± 0.058 against a predicted 2.500, the minimum landing exactly on prediction for 10 of 12 models and a wrong-sign control beaten 12 of 12.

Why this matters for interpretability. The immediate consequence is that a probe can be perfect and still wrong. We fit a polynomial readout to true physical energy; it reaches $|\rho| = 0.9999$, as good as a probe can be. Yet it is $6.7\times$ less preserved by the model’s own

transition than the label-free scalar, and enforcing it during imagination makes the model’s physics worse, while the label-free scalar improves it by 42%. Nothing is wrong with the probe. The probe is fitted to a quantity the generating process does not conserve — and no amount of probe hygiene (Hewitt and Liang, 2019; Belinkov, 2022) detects that, because probe hygiene tests the probe.

Measuring the operator instead. The discriminating measurement is not on the representation but on the transition: apply one autonomous step at a real encoded state and ask how much a candidate scalar moves (Equation 2). This single statistic separates trained from untrained models ($74\times$), conservative from dissipative ($39\times$, no overlap), in-distribution from out ($55\text{--}267\times$), one architecture from another ($767\times$), and a label-free scalar from a supervised probe of the same quantity ($6.7\times$). On every one of those contrasts a probe reports success. The dissociation is directional in each case we can construct: decodability is retained where dynamical structure fails.

The quantity the operator selects is used by the model. Projecting the latent back toward its level set during imagination reduces drift in decoded physical energy by 55–76% at long horizon on trajectories the method never saw, with 0 of 60 magnitude-matched random directions and 0 of 15 equal-norm tangent directions beating it. Setting the scalar to an independent trajectory’s value moves the imagined world’s true energy toward that trajectory’s (rank correlation 0.80–0.91, 0 of 75 controls). The same interchange applied 50 steps into an autonomous rollout — on a state the model reached by itself — works nearly as well, which a dormant pathway would not (Makelov et al., 2024).

Contributions. (i) World models learn their simulator’s numerical scheme: the recovered invariant is the integrator’s shadow Hamiltonian, and its coefficient tracks the simulator timestep across a $4\times$ range with a parameter-free slope of 2.484 ± 0.058 against 2.500. (ii) A demonstration that probing over-reports, with a decisive control and a mechanism: a supervised probe of the true quantity, at $|\rho| = 0.9999$, identifies a direction the dynamics do not use, because it is fitted to a quantity the generating process does not conserve; the dissociation appears on five independent axes and is directional in all of them. (iii) An operator statistic that discriminates where probes cannot, with magnitude-matched rather than norm-matched controls — the norm-matched null used in prior work on this system controls nothing, because the correction is scale-invariant in the constraint. (iv) Causal evidence and stated boundaries: the selected direction is deployed, including 50 steps into autonomous imagination, and the correction confers no benefit to a planner, for a reason we measure.

2 Recovering a latent invariant

Model and data. We analyze the published DreamerV3 architecture (Hafner et al., 2023): 13.5M parameters, categorical 32×32 stochastic latents, a convolutional encoder and decoder, KL balancing, and unimix. We train only the world model, with no actor or critic.

Data come from Gymnasium’s Pendulum-v1 (Towers et al., 2024), rendered per step and block-averaged to 64×64 , with the simulator state set directly ($\theta \sim U[-1.8, 1.8]$, $\dot{\theta} \sim U[-2.2, 2.2]$) and any trajectory reaching the $|\dot{\theta}| = 8$ clip rejected, since clipping breaks conservation. Actions are zero unless stated, and a single frame does not reveal velocity, so the model must infer $\dot{\theta}$ from the sequence. We train on 204 trajectories of 120 frames for about 6,500 gradient steps, reserving 52 as a post-training analysis set. Three seeds pass checks fixed before analysis: KL above 1 nat, one-step decoding $4\times$ better than the dataset mean, and finite rollouts; Appendix A gives the rest. Our claims concern a DreamerV3 world model trained on pendulum video, not DreamerV3 as a full reinforcement-learning agent.

Latent coordinates. After training we freeze the model. Let h be the deterministic recurrent state on the analysis set and $\bar{h}$ its mean. We take the top 12 principal directions of h , drop any without support in the data, and let P be the resulting orthonormal map. All extraction

and perturbation operate on $z = P^\top(h - \bar{h})$. Let $F(z)$ be the projection through P of the model’s one-step displacement $T(h) - h$; using the model’s own displacement avoids fitting a surrogate vector field and then analysing that instead.

The candidate family. We search degree-4 polynomials in z : with $\varphi(z)$ the monomials up to degree 4 in the 12 latent coordinates, a candidate is $C(z) = a^\top \varphi(z)$ over 1819 monomials, excluding the trivially conserved constant. Energy is quadratic in $\dot{\theta}$ and needs polynomial terms to approximate $\cos \theta$, and the latent is an unknown nonlinear encoding of $(\theta, \dot{\theta})$ rather than those coordinates, so the family must express energy through that distortion while staying small enough to fit.

The invariance criterion. An invariant is constant along a trajectory and differs between trajectories, so we score a candidate by

$$\text{ratio}(C) = \frac{\text{mean within-trajectory variance of } C}{\text{total variance of } C}, \quad (1)$$

which is 0 for a perfectly conserved scalar and near 1 for one wandering as much within a trajectory as across the dataset. The denominator keeps it non-trivial: without it $C = 0$ would score perfectly while distinguishing nothing. Both variances are quadratic forms in a , so with W the mean within-trajectory covariance and T the total covariance this is a Rayleigh quotient $a^\top W a / a^\top T a$, minimised by the smallest generalized eigenvector of $W a = \lambda T a$. One eigendecomposition returns the whole family, ranked.

Selecting among conserved candidates. Conservation alone does not pick out a unique scalar: if C is conserved so is any function of it, so E , E^2 and mixtures all sit near the top and the leading eigenvector is generally a combination. We therefore fit C jointly with an antisymmetric operator B such that $F \approx B \nabla C$, $B^\top = -B$, mirroring the Hamiltonian relation between a conserved quantity and the flow it generates (Arnold, 1989). Since $\nabla C^\top B \nabla C = 0$ for antisymmetric B , such a scalar is constant along the flow by construction, and the criterion separates candidates the ratio cannot — E^2 is exactly as conserved as E but does not pair with the same B . The fit is bilinear, so we alternate least-squares solves, searching within the top eight eigenvectors so the recovered C has eight effective degrees of freedom rather than 1819 — which matters on 52 trajectories, where a free fit could drive the in-sample ratio to zero by overfitting. Appendix A isolates its contribution: conservation drives most of the recovery and intervention effect, while flow alignment mainly reduces variation across seeds.

Reference energy. We compare C against textbook pendulum energy $E = \frac{1}{6}\dot{\theta}^2 + 5 \cos \theta$. Gymnasium integrates semi-implicitly, so it does not conserve E exactly; Section 4 shows what it conserves instead, and that this is the crux rather than a caveat. The search sees only latent states and the model’s own one-step transition — never θ , $\dot{\theta}$ or E . Ground truth enters afterward, when scoring C against E . The extraction dimension was selected on three development models and fixed before evaluating the three reported here.

3 Decodability is not dynamical structure

A probe establishes that a quantity can be read out of a representation. It says nothing about whether the model’s own transition uses or preserves that quantity. We measure the second property directly, with a statistic on the transition operator: apply one autonomous step at a real encoded state and ask how much a candidate scalar changes,

$$\rho_{\text{obs}}(C) = \frac{\text{median}_t |C(T(z_t)) - C(z_t)|}{\text{std}_{\text{traj}} C}, \quad (2)$$

normalised so it is invariant to the scale of C and comparable across candidates. A value near zero means the transition preserves C ; a value near one means a single step moves C by its entire across-trajectory spread.

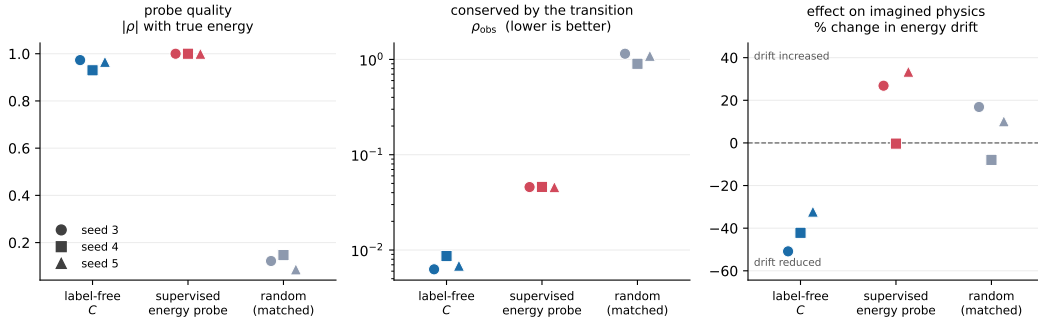


Figure 1: A perfect probe is not a conserved quantity. Colour is the constraint; marker is the model seed (all three shown, never averaged). Left: the supervised probe reads true energy almost perfectly. Centre: the model’s own transition preserves it an order of magnitude worse than the label-free scalar. Right: acting on it fails to reduce rollout energy drift, while the label-free scalar cuts drift on every seed. Probe quality and dynamical relevance rank the three constraints differently.

Table 1: Decodability is retained while dynamical structure degrades, across five independent axes. ρ_{obs} for the trained conservative model in-distribution is $6.3\text{--}8.7 \times 10^{-3}$.

axis	contrast	decodability	ρ_{obs} degradation
training	untrained network	$ \rho _E$ up to 0.908	$74\times$
state space	outside training energies	free probe 0.999	$55\text{--}267\times$
architecture	deterministic conv-GRU	$ \rho _E$ 0.91 (2 of 3 seeds)	$767\times$
objective	transition never trained	$ \rho _E$ 0.97	$\sim 10\,000\times$
actuation	torque applied	$ \rho _E$ 0.72–0.88	balance law absent

A perfect probe identifies a direction the dynamics do not use. The sharpest test replaces the label-free search with the answer it is trying to find. We fit a degree-4 polynomial readout to true pendulum energy by ridge regression on the same latent, then use it to drive the identical correction. It reaches $|\rho|_E = 0.9999$: as a probe it is essentially perfect.

Figure 1 shows the outcome. The supervised probe tracks energy better than the label-free scalar on every seed, yet is $6.7\times$ less preserved by the transition — median ρ_{obs} of 4.58×10^{-2} against 6.9×10^{-3} , and $5.3\text{--}7.3\times$ per seed; both hold on 3 of 3 models. It also never repairs the rollout: its effect on energy drift is $+26.8\%$ and $+33.4\%$ on two models and -0.3% on the third, against a median -42.2% ($32\text{--}51\%$) reduction from the label-free scalar on all three. A magnitude-matched random constraint sits at $\rho_{\text{obs}} \approx 1.07$ and changes drift by $+4.0\%$. The direction that best predicts the physical variable and the direction the model’s dynamics actually respect are not the same direction.

This also answers the practical objection to a label-free method. The search is not a workaround for missing labels: it finds something a supervised fit on perfect labels does not, because it optimises conservation by the transition rather than agreement with the variable. Section 4 identifies what it finds, and why fitting true energy cannot find it.

The dissociation is directional, and appears on five further axes. In every case we can construct, decodability survives where conservation fails; we have not found the converse. Table 1 collects them. On each row a probe would report success.

The out-of-distribution row is cleanest: it holds the model fixed and varies only the region of state space. Outside the training energy band a freely fitted probe still recovers energy at 0.999 — the encoder generalises — while the transition conserves it $55\text{--}267\times$ worse, reaching values indistinguishable from a random constraint. The two properties come apart inside a single trained model, split by where in state space it is asked.

The actuation row is the only axis testing a balance law rather than a conservation law; Appendix B gives it in full.

Whether conservation alone predicts the intervention is not established. Among candidates matched in energy correlation and spanning five orders of magnitude of conservation, the rank correlation with repair is +0.19, +0.57 and +0.19 across seeds, with the 95% interval excluding zero on one of three. We report this as a negative, and note why the control is hard: on a system whose transition conserves energy the well-conserved latent directions are the energy-like ones, so a matched band exists only at low decodability and on one seed cannot be built at all. Appendix B gives the full analysis. What holds on all three seeds is that the recovered scalar repairs: -16.7% , -7.3% and -6.8% .

Where the phenomenon stops. Both boundaries are the same statement twice: the invariant is bounded to the training distribution and to RSSM-like architectures. On a deterministic conv-GRU, identification transfers on 2 of 3 seeds under both training budgets while conservation transfers on none — a $767\times$ gap — and seven times more open-loop training moves the median barely. That comparison does not isolate architecture, since an open-loop reconstruction term and a prior-posterior KL are different losses; Appendix B gives both boundaries in full.

4 Why the perfect probe fails

Section 3 shows the optimal readout of true energy is a poor invariant of the model’s transition, but not why. The reason is that the probe is fitted to the wrong quantity.

The data does not conserve textbook energy. Trajectories come from semi-implicit (symplectic) Euler,

$$\dot{\theta}_{n+1} = \dot{\theta}_n + \frac{3g}{2l} \sin(\theta_n) \Delta t, \quad \theta_{n+1} = \theta_n + \dot{\theta}_{n+1} \Delta t. \quad (3)$$

A symplectic integrator conserves not the continuous system’s Hamiltonian H but a nearby shadow $\tilde{H} = H + \mathcal{O}(\Delta t)$ (Hairer et al., 2006a). Here the first-order correction is $\frac{\Delta t}{2} \{T, V\} \propto \dot{\theta} \sin \theta$, giving

$$\tilde{H} = E + c^* \dot{\theta} \sin \theta, \quad c^* = \frac{\Delta t}{2} mg \frac{l}{2} = 0.125, \quad (4)$$

with no free parameters. If the model learned the integrator’s map rather than the textbook physics, $\tilde{H}$ is what its transition should preserve, and a probe fitted to E is fitting the wrong target however well it fits.

A sweep, not a two-point test. Comparing a probe fitted to E against one fitted to $\tilde{H}$ would not settle this: $\tilde{H}$ is a different function and might be better conserved merely because it is easier to represent. We therefore sweep the family

$$T_c(\theta, \dot{\theta}) = E(\theta, \dot{\theta}) + c \dot{\theta} \sin \theta, \quad (5)$$

over a grid containing c^* , both signs, and wrong magnitudes $2\times$ and $4\times$ either way, refitting at each c with everything else fixed. At $c = 0$ this is exactly the supervised probe of Section 3, reproducing its numbers to five decimals. The prediction is then the location of a minimum, which no representational-ease artifact produces, and it was registered before the sweep ran.

The minimum is where the integrator says it is. Figure 2 shows both halves. On ground-truth trajectories, with no model involved, the invariance ratio of T_c falls by $591\times$ between $c = 0$ and $c = c^*$, confirming the derivation and fixing the sign. On the learned transition, ρ_{obs} is minimised at c^* on 3 of 3 models, falling $4.9\text{--}6.9\times$ below its value at $c = 0$. The curve is not monotone in $|c|$: the wrong-sign, right-magnitude control at $c = -c^*$ is $12\times$ worse than at $c = +c^*$ (range $10\text{--}14\times$ over the three models). A post-hoc refinement over $c \in [0.090, 0.170]$ locates the minimum at 0.125 ± 0.005 on every seed.

The repair follows the same curve. Enforcing T_{c^*} during imagination changes energy drift by -49.5% , -31.5% and -29.3% on the three models — a repair on all three — where enforcing T_0 , the perfect probe, gives $+26.8\%$, -0.3% and $+33.4\%$. Correcting an $\mathcal{O}(\Delta t)$ term in the target flips the intervention from harmful to helpful.

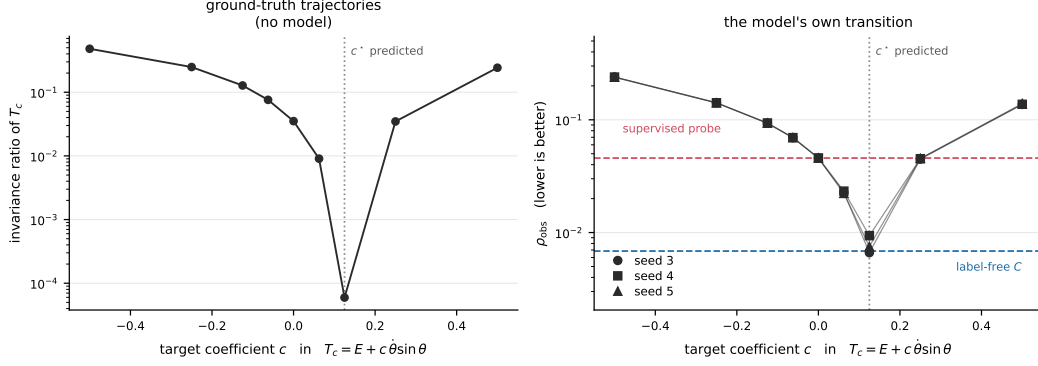


Figure 2: The model’s transition conserves the integrator’s shadow Hamiltonian, not textbook energy. Both panels sweep the target $T_c = E + c\theta \sin \theta$; the dotted line is $c^* = 0.125$ predicted from the integrator with no fitting. Left: on ground-truth trajectories, with no model involved, T_{c^*} is $591\times$ better conserved than textbook energy. Right: the learned transition agrees, on all three seeds. The sweep passes from the supervised probe’s level (red) down to the label-free scalar’s (blue) exactly at c^* . The wrong-sign control at $c = -c^*$ is $12\times$ worse (median over seeds), so this is not monotone in $|c|$.

Table 2: The recovered correction tracks the simulator’s timestep across a $4\times$ range. Three models per timestep; ρ_{obs} is the median across seeds.

Δt	predicted c^*	$\arg \min_r$ per seed	ρ_{obs} at $r=0$	at $r=1$	separation
0.02	0.050	+0.75, +0.75, +1.00	0.0171	0.0166	$1.03\times$
0.035	0.088	$+1.00 \times 3$	0.0231	0.0103	$2.24\times$
0.05	0.125	$+1.00 \times 3$	0.0449	0.0079	$5.72\times$
0.08	0.200	$+1.00 \times 3$	0.1188	0.0088	$13.5\times$

The model tracks the simulator’s timestep. If the model has learned the integrator rather than the physics, then c^* is not a constant of the pendulum but a property of the discretisation, and changing the simulator’s timestep should move it: $c^*(\Delta t) = \frac{\Delta t}{2} mg \frac{l}{2} = 2.5 \Delta t$. We test this directly. Four datasets are generated at $\Delta t \in \{0.02, 0.035, 0.05, 0.08\}$, identical in every other respect, and three models are trained on each. The sweep is expressed relatively, $r = c/2.5\Delta t$, so the prediction is a single value of r at every timestep.

Table 2 shows the outcome. The minimum sits exactly at the predicted $r = 1$ on 10 of 12 models, and the wrong-sign control is beaten on 12 of 12. Regressing the recovered coefficient on Δt through the origin, as the theory requires, gives a slope of 2.484 ± 0.058 against a parameter-free prediction of 2.500 — agreement to 0.6% with nothing fitted.

Two scalings make the mechanism visible. As the timestep coarsens, ρ_{obs} for textbook energy grows as $\Delta t^{1.41}$, while ρ_{obs} for the shadow quantity stays at the model’s own floor, 0.008–0.017 across the whole range. The separation between them therefore grows from $1.03\times$ to $13.5\times$ purely because textbook energy becomes worse conserved while the shadow quantity does not. At every timestep, the model conserves the integrator’s quantity rather than the physicist’s.

We report one registered failure. Our preregistration required both a slope within 20% and an intercept whose interval contains zero; the two-parameter fit gives a slope of 2.611 ± 0.110 but an intercept of -0.0072 ± 0.0057 , which excludes zero. It is driven entirely by the smallest timestep, where a post-hoc finer grid puts the minimum at $r = 0.875$ on all three seeds — a real 12% low bias rather than grid coarseness, in the regime where the whole separation is only $1.03\times$ and the residual curve is flat to within 2%.

What this does and does not settle. The residual matters and we report it: median ρ_{obs} at c^* is 7.5×10^{-3} against the label-free scalar’s 6.9×10^{-3} , a gap of $1.10\times$ — so the shadow

Hamiltonian accounts for essentially all of the $6.7\times$ gap in Section 3. It does not follow that the model conserves $\tilde{H}$ exactly: it is a learned approximation to the integrator, so its own invariant is its own modified quantity, of which $\tilde{H}$ is a first-order proxy. That $1.10\times$ measures the difference.

Two consequences follow. A probe reaching $|\rho|_E = 0.9999$ can be fitted to a quantity the generating process does not conserve, and no probe hygiene detects this, because nothing is wrong with the probe — the target is. And the label-free method’s advantage has a name: it optimises conservation under the learned transition, so it is not restricted to targets a human knows how to write down, and here it recovered an $\mathcal{O}(\Delta t)$ correction nobody supplied. The thesis is internal to a single sweep: at c^* the probe correlates with textbook energy worse than at $c = 0$ (0.977 against 0.9999) while being conserved six times better and repairing the rollout instead of harming it.

5 The privileged direction is causally deployed

If the recovered scalar were only a descriptive correlate, enforcing it would have no reason to help and setting it no reason to steer. Three interventions say otherwise, each against matched controls that differ only in where they push, never in how hard — a distinction that matters, since the obvious norm-matched null controls nothing (Appendix B).

Evaluated on decoded physical energy rather than pixels, projecting the latent back toward its level set during imagination reduces drift by 55–76% at $H = 190$ on 512 trajectories from a disjoint generator seed, with the scalar and coordinate frame frozen — and 0 of 60 magnitude-matched random directions and 0 of 15 equal-norm tangent directions beat it. Setting the scalar to an independent donor trajectory’s value moves the imagined world’s true decoded energy toward that trajectory’s true energy (rank correlation **0.802–0.914**, 0 of 75 controls), with a preregistered offset sweep monotone in both directions on every model. Repeating that interchange 50 steps into an autonomous rollout — on a state the model reached by itself, not one the encoder produced — costs little (0.762–0.849 against 0.803–0.924 at depth 0, 0 of 13 controls), which a dormant pathway would not survive (Makelov et al., 2024).

The same pipeline refuses when there is nothing to find. On matched dissipative models, where no nontrivial first integral exists, it returns a direction indistinguishable from random: 24 of 60 random draws beat it, and its operator statistic sits at $3.4\text{--}3.7 \times 10^{-1}$ — the range of an untrained network, $39\times$ worse than any conservative model, with no overlap. Appendix B gives protocols and per-model numbers.

6 Limitations

Two smooth Hamiltonian systems — a pendulum and a two-degree-of-freedom anharmonic oscillator — with no contacts and no multiple interacting objects; whether any of this survives discontinuous dynamics is untested. Three trained models per arm, with the model seed as the independent unit and ranges rather than confidence intervals, because three is too few for a precise effect size. The extraction minimises an invariance ratio that is the normal component of one-step error, so a low normal error is the objective being met, not a finding; the empirical content is that the resulting direction coincides with physical energy, which supervision does not reproduce. The architecture comparison likewise does not isolate architecture: an open-loop reconstruction term and a prior–posterior KL are different losses, not the same loss at different strengths.

No downstream benefit, and we measured why. Both practical uses we tested fail, for one reason. Planning with CEM over the learned model on an energy-targeting task, the model plans well — beating random actions by +1.65, 95% CI [+0.50, +2.79] — yet no correction arm differs from no correction over three models and 20 episodes each: the largest effect is 0.0014 against a return standard deviation of 0.575, 0.24% of one episode’s spread, with all four registered predictions failing. The cause is structural. The correction acts on accumulated drift, and at the planner’s horizon it shifts imagined energy by 0.3% of the

spread across candidate action sequences — the quantity the planner ranks plans by — so it cannot change which plan is chosen. Our repair is measured over 100 autonomous steps; a re-planning controller never imagines long enough to accumulate what the method removes, and looking further ahead does not help, because planning quality itself degrades with horizon. The same effect-size limit defeats drift as an online trust signal, which is beaten by plain latent displacement on every model. Appendix B gives both in full.

The method finds only what the data gave the model a reason to represent. The pendulum’s rod length is fixed, an exact constraint, yet searching for a residual $G(z) = 0$ fails: the recovered direction correlates with the true residual at 0.004–0.032. Minimising $\mathbb{E}[G^2]$ is the wrong objective, ranking the true constraint 1525th of 1819 directions, and the constraint is barely encoded at all — predicted from the latent at held-out $|\rho| = 0.19$. A quantity that never varies carries no information, so nothing in a reconstruction objective pushes it into the latent; it can live in the decoder’s weights. This bounds the approach to quantities that vary, and is why we report no released-checkpoint experiment: limb lengths do not vary either.

Finally, the two-invariant case limits the extraction rather than the model. Where both energy and angular momentum are conserved, the conserved subspace reliably contains both (angular-momentum correlation 0.94–0.98) while the joint fit isolates either in only 1 of 7 measurements: flow alignment selects a single direction inside that subspace and has no unique answer when two exist.

7 Related work

What probes establish. Probing shows a variable is recoverable, not that the computation uses it (Alain and Bengio, 2017). Control tasks (Hewitt and Liang, 2019) and later critiques (Belinkov, 2022) address the confound between what the representation stores and what the probe can learn. Ours is a different failure: the probe in Section 3 is fitted to the true quantity and reaches 0.9999, so no control-task correction applies, and it still identifies a direction the dynamics do not use. Work on emergent world representations goes beyond probing by intervening (Li et al., 2023; Nanda et al., 2023), the logic ours follows; Makelov et al. (2024) show such interventions can succeed through pathways the model does not normally use, which our depth-50 interchange answers.

Discovering conserved quantities. ConservNet minimises a noise-regularised within-group variance (Ha and Jeong, 2021), of which our invariance ratio is the closed-form Rayleigh-quotient analogue; AI Poincaré counts conserved quantities from trajectory manifolds (Liu and Tegmark, 2021). ConCerNet learns a conservation law then projects the learned dynamics onto it (Zhang et al., 2023), FINDE finds first integrals and preserves them by projection (Matsubara and Yaguchi, 2023), and Noether Networks meta-learn a conserved quantity and apply it at test time, including on pixel pendulums (Alet et al., 2021). All impose or co-train conservation. We ask whether a model trained with no such term already has a scalar its transition preserves, measured on the frozen transition rather than built in.

Structure by design, and whether it is what helps. Hamiltonian and Lagrangian networks build conservation into an architecture (Greydanus et al., 2019; Cranmer et al., 2020). Gruver et al. (2022) find HNN generalisation comes from modelling acceleration and coordinate choice rather than from symplectic structure, which licenses the objection that our projection is a generic regulariser. The magnitude-matched null answers it: with step size fixed, 0 of 60 random and 0 of 15 equal-norm tangent directions match the recovered one, and on dissipative models the pipeline returns a direction indistinguishable from random.

Drift, and evaluating physics in world models. That accurate one-step models drift over long rollouts is documented across fields; in machine-learned force fields, force accuracy and simulation stability decouple sharply (Fu et al., 2023). Projection onto a level set is a standard structure-preserving device (Hairer et al., 2006b), and our correction applies it to a discovered rather than a given invariant, in a learned latent. World models are

otherwise evaluated by prediction quality or control performance (Hafner et al., 2023; Ha and Schmidhuber, 2018; Schrittwieser et al., 2020), and physics benchmarks evaluate generated output — neither reads the internal state. The disentanglement literature (Locatello et al., 2019) documents how readily an appealing structural claim about representations survives without the control that would falsify it.

8 Conclusion

A world model trained only on pendulum video learns a conserved quantity, and it is the one its simulator conserves rather than the one physics prescribes. Vary the simulator’s timestep and the recovered coefficient follows the integrator’s parameter-free prediction, slope 2.484 ± 0.058 against 2.500. What the model has internalised is the numerical scheme, not the physics that scheme approximates.

That reframes a failure of interpretability rather than excusing it. A probe fitted to true energy at $|\rho| = 0.9999$ identifies a direction the dynamics do not use, and enforcing it degrades the model’s imagined physics — not because the probe is poor, but because it is fitted to a quantity the generating process does not conserve. No probe hygiene detects that. Decodability and dynamical structure are different properties, they come apart in every setting we could construct, and only a measurement on the transition operator tells them apart.

The practical consequence is narrow. That a physical variable is decodable says little about whether the model’s rollouts respect it, and says least where it matters most: out of distribution, in architectures that predict well without conserving, and when the variable comes from supervision rather than from the dynamics. The operator statistic costs one autonomous step per state; it does not, on this evidence, buy a better controller.

A Experimental details and extraction ablations

A magnitude-matched null, and why the obvious one fails. The natural correction projects the latent back toward the level set, $z \leftarrow z - \alpha(C(z) - C_0) \nabla C / \|\nabla C\|^2$. This is a Newton step, so its size grows with how badly the constraint is violated — and a random constraint is violated far more than a conserved one. Matching random draws in coefficient norm, as one might expect to control for scale, controls nothing: the step is invariant under $C \rightarrow \lambda C$. Measured, random draws take steps $29\times$ larger than the recovered constraint, so the comparison conflates direction with magnitude.

We therefore fix the step size and vary only direction, $z \leftarrow z - \varepsilon \text{sign}(C - C_0) \nabla C / \|\nabla C\|$, over a frozen ε grid. Arms then differ in where they push, never in how hard.

Three kinds of latent trajectories. We use three latent trajectories and keep them separate throughout.

An observation-conditioned trajectory is the sequence of deterministic recurrent states produced when the model receives the next real frame at every step. This is what the encoder returns on the analysis set.

The one-step transition T is the model’s autonomous recurrent update with zero action and no new observation.

An imagination rollout starts from one encoded state and repeatedly applies T without supplying any later frames.

We estimate the latent flow by applying T once at each state of an observation-conditioned trajectory, so the transition is autonomous even though the states come from real video. Section 3 measures conservation along observation-conditioned trajectories. Section 5 tests what happens to the same scalar under autonomous imagination.

Code and data. The extraction code, the run logs behind every figure and table, and the source of this paper are at <https://github.com/Zarand3r/world-model-invariants>. The figures regenerate from the committed logs without a GPU.

Preliminary experiments. We developed the extraction procedure in preliminary experiments on smaller recurrent models. Those experiments motivated the untrained, dissipative, and intervention controls used here, and we do not use them as evidence for the DreamerV3 results.

Training. We set free bits to 0 rather than DreamerV3’s reference default of 1.0. The measured KL divergence settles near 1.8 nats, well above collapse, so the floor is inactive here. All three trained models pass the acceptance checks; none were discarded.

Data split. Training uses trajectories 0–203 of the 256-trajectory dataset. The remaining 52 form the analysis set. We fit C on those 52 trajectories and evaluate the Section 5 correction on the same 52, so the absolute effect is in-sample with respect to invariant fitting. The recovered and random-constraint arms use the same trajectories.

Scoring the correction over a grid. We score the slope of rollout error over the fixed α grid rather than the minimum on that grid. Taking the minimum selects the most favourable of five points after seeing the curve. In one diagnostic run that rule would have reported a 5.9% improvement for an arm whose error rose monotonically to twice baseline.

Flow alignment contributes reproducibility. Holding the candidate family, data, dimension, degree, and α grid fixed, we select C either by the invariance criterion alone or by the joint criterion. Conservation alone recovers energy at 0.950 against 0.973 for the joint rule, and gives a slightly better mean intervention slope (-0.087 against -0.077 , winning on two of three models). The difference is in spread: recovery varies by 0.035 across seeds under conservation alone and by 0.008 under the joint rule.

The pairing residual does not track recovery. As the extraction dimension rises from 6 to 12, energy recovery improves from 0.721 to 0.967 while the pairing residual worsens from 0.738 to 0.876. Recovery already reaches 0.967 at dimension 8, where the residual is lower at 0.813. The added dimensions do not carry energy while resisting the Hamiltonian fit: directions 7–12 account for 44.9% of the residual and 43.5% of the flow. Figure 4 gives the sweep.

B Causal-intervention protocols and per-model numbers

If the recovered scalar is only a descriptive correlate, enforcing it has no reason to help and setting it has no reason to steer. Three interventions test the alternative, each with matched controls.

A magnitude-matched null. Arms differ only in where they push, never in how hard: we fix the step size and vary direction, $z \leftarrow z - \varepsilon \text{sign}(C - C_0) \nabla C / \|\nabla C\|$, over a frozen ε grid. The obvious alternative — a level-set projection with random draws matched in coefficient norm — controls nothing, because the step is invariant under $C \rightarrow \lambda C$; measured, random draws then take steps $29\times$ larger than the recovered constraint. Appendix A gives the derivation.

Correcting the invariant repairs the imagined physics. We evaluate on decoded physical energy rather than pixels: a non-learned geometric readout recovers θ by image moments and $\dot{\theta}$ by backward differences, matching the simulator’s semi-implicit convention, with median decoded-energy error 1.6% of the across-trajectory spread. The statistic is secular drift of decoded energy over a rollout.

At $H = 100$ the recovered direction reduces drift by 32–51% across three models. On 512 trajectories from a disjoint generator seed, with C and the entire coordinate frame frozen, the effect does not shrink: -54.6% , -58.8% , -75.9% at $H = 190$. Across both horizons 0

of 60 magnitude-matched random directions and 0 of 15 equal-norm tangent directions beat the recovered one on any model.

Setting the invariant steers the imagined physics. Repair shows drift is costly, not that C is a control variable. We move a recipient state minimally along the local C -normal until $C(z) = C(z_{\text{donor}})$ for an independent donor trajectory, then let the rollout run free; the target is read from the model’s own latent, never from true energy. The imagined world’s true decoded energy moves toward the donor’s true energy, rank correlation 0.802–0.914 across three models, with 0 of 75 random and tangent controls beating it and a preregistered offset sweep monotone in both directions on every model.

The subspace is used by the model’s own dynamics, not only reachable from encoder states. A subspace edit can produce the expected output change through a pathway the model does not normally use (Makelov et al., 2024), so we repeat the donor interchange 50 steps into an autonomous rollout — on a state the model reached by itself. The effect barely degrades: rank correlation 0.762–0.849 at depth 50 against 0.803–0.924 at depth 0, with 0 of 13 controls beating it at either depth, and equal-norm tangent edits near zero throughout. A dormant pathway would lose its effect precisely where this test applies it.

The same pipeline refuses when there is nothing to find. On matched dissipative models, where no nontrivial first integral exists, the extraction returns a direction that is not distinguished from a random one: 24 of 60 random draws beat it, and it harms rollouts on 2 of 3 models. Its operator statistic sits at $3.4\text{--}3.7 \times 10^{-1}$, in the same range as an untrained network and $39\times$ worse than any conservative model, with no overlap between the two groups.

Downstream tests in full.

No downstream benefit, and we measured why. Both practical uses we tested fail, for one reason. Planning with CEM over the learned model on an energy-targeting task, scoring return from the simulator’s true state, the model plans well — beating a random-action policy by +1.65, 95% CI [+0.50, +2.79]. But no correction arm, including one supplied with the action’s power as a source term, differs from no correction over three models and 20 episodes each: the largest effect is 0.0014 against an across-episode return standard deviation of 0.575, 0.24% of one episode’s spread, and all four registered predictions failed. The cause is structural. The correction acts on accumulated drift, and at the planner’s horizon it shifts imagined energy by 0.3% of the spread across candidate action sequences — the quantity the planner ranks plans by — so it cannot change which plan is chosen. Our repair is measured over 100 autonomous steps; a controller that re-plans every step never imagines long enough to accumulate what the method removes. Nor can one simply look further ahead: tripling the horizon leaves every arm indistinguishable (largest effect 0.007 against a return standard deviation of 0.862) while planning quality itself degrades, -0.34 to -0.56 in mean return. The regime where the correction would have something to correct is the regime where the planner is already worse off.

The same effect-size limit defeats the other use. Accumulated drift at rollout step 25 predicts decoded energy error at step 100 at Spearman -0.02 , $+0.09$ and -0.24 , against $+0.27$, $+0.29$ and $+0.31$ for plain latent displacement, with a random-constraint control beating it on two of three; the registered prediction went the other way on every model. A conserved quantity being causally deployed in imagination does not imply its drift is a useful online signal. The effect on pixel error alone is also small (1.7% at $H=50$); the large effects here are on decoded physical energy, a metric we introduce because it measures the property in question.

The method finds only what the data gave the model a reason to represent. The pendulum’s rod length is fixed, so the rendered bob’s distance from the pivot is constant to 0.67% — an exact constraint. Searching for a residual $G(z) = 0$ rather than a conserved scalar fails: the recovered direction correlates with the true residual at 0.004–0.032. Two measured causes. Minimising $\mathbb{E}[G^2]$ is the wrong objective, ranking the true constraint 1525th of 1819 directions; and the constraint is barely encoded at all, predicted from the latent at held-out

$|\rho| = 0.19$ with a residual four times its own variation. A quantity that never varies carries no information, so nothing in a reconstruction objective pushes it into the latent — it can live in the decoder’s weights. This bounds the approach to quantities that vary, and it is why we report no released-checkpoint experiment: limb lengths do not vary either.

Finally, the two-invariant case limits the extraction rather than the model. On a system conserving both energy and angular momentum the conserved subspace reliably contains both (angular-momentum correlation 0.94–0.98 in every measurement), while the joint fit isolates either in only 1 of 7 measurements. The flow-alignment criterion selects a single direction inside that subspace, and has no unique answer when two independent invariants exist.

Conservation at matched decodability, in full.

Whether conservation alone predicts the intervention is not established. The control that would show this directly asks whether, among candidates matched in energy correlation, better conservation alone produces a larger repair. The 2-DoF system admits such a band: on two of three seeds, 397 and 398 of 400 candidates lie within 0.10 of the recovered scalar’s $|\rho|_E$, spanning five orders of magnitude of conservation quality at $|\rho|_E \leq 0.11$. Across 20 candidates stratified by conservation, the rank correlation with repair is +0.19, +0.57 and +0.19 on the three seeds, with the 95% confidence interval excluding zero on one of three. We report this as a negative result: the relationship is visible but not established at $n = 3$.

Two further facts belong with it. The point estimate moves with the size of the candidate pool ($+0.29 \rightarrow +0.19$ on one seed between pools of 150 and 400), so the statistic is not stable to a design choice that should be immaterial — though the one-of-three verdict is identical under both pools. And on the third seed the band cannot be built at all: its reference $|\rho|_E$ is 0.111 and only 19 of 400 candidates fall within 0.10 of it, with an increase from 150 to 400 candidates adding none.

That last failure is the informative part. On a system whose transition conserves energy, the well-conserved latent directions are the energy-like ones. This is why E10’s original high-decodability form could not be constructed on the pendulum at any pool size, why the matched band here exists only at low decodability, and why on one seed it does not exist at all. The two properties may not be separable by this design on any system where the model has learned the physics. That is a statement about the difficulty of this control, not about the dissociation itself, for which the supervised probe above and Section 4 give direct evidence. What does hold across all three seeds is that the recovered scalar repairs: -16.7% , -7.3% and -6.8% .

Under actuation the same split appears against a different law. The four axes above vary how well a conserved quantity survives; a fifth varies the law itself. Three further models are trained on the same pendulum under piecewise-constant random torque, where energy instead obeys $dE/dt = \tau\dot{\theta}$. They demonstrably use their actions: a 20-step open-loop rollout under the true action sequence beats one under the same actions shuffled across trajectories, 0.740–0.761 on 3 of 3 seeds. The unchanged extraction recovers an energy-like scalar on every model ($|\rho|_E = 0.72, 0.88, 0.86$), yet its motion under the model’s own transition does not follow the balance relation: with the torque known and $\dot{\theta}$ from the same decoded readout, the rank correlation with the predicted source term is 0.041, 0.076 and 0.065, the power term accounts for 0.31% of the variance in ΔC , and the observed change is $4.5\text{--}5.4\times$ larger than predicted. A shuffled-power control is beaten 3 of 3, so the pairing carries something — almost nothing. The model represents the quantity and does not implement the relation it obeys, which is also why a correction built on that scalar confers no control benefit (Section 6): there is no balance structure for it to enforce.

Where the phenomenon stops. Both boundaries above are the same statement seen twice. The invariant is bounded to the training distribution: outside the training energy band a freely fitted probe still recovers energy at 0.999 while the transition conserves it $55\text{--}267\times$ worse, reaching values indistinguishable from a random constraint. And it is bounded to the architecture: on a deterministic convolutional GRU with no stochastic latent, no KL term

and no unimix, identification transfers on 2 of 3 seeds under both training budgets while conservation transfers on none (ρ_{obs} 4.83–5.55 against the RSSM’s $\sim 7 \times 10^{-3}$, a $767\times$ gap; seven times more open-loop training moved the median from 5.33 to 5.26). We are explicit that this comparison does not isolate architecture — an open-loop reconstruction term and a prior–posterior KL are different losses, not the same loss at different strengths — so the supportable scope of every claim here is RSSM-like models, in distribution.

C Latent sensitivity analyses

For each direction u of the extracted subspace we measure its variance and the damage a displacement along it does to an imagination rollout,

$$V(u) = \text{Var}(u^\top z), \quad D_H(u) = \mathbb{E}[L_H(z + \epsilon u) - L_H(z)], \quad (6)$$

with L_H the H -step imagination error against the analysis set.

Calibrating the displacement. At $\epsilon = 0.05|z|$ the damage is about -0.5% of baseline with the sign varying across directions. Imagination is deterministic, so repeated unperturbed rollouts are identical and these values are not sampling noise. Damage for the leading direction runs -0.5% , $+3.1\%$, $+18.1\%$, $+46.4\%$, and $+128.8\%$ at $\epsilon/|z|$ of 0.05, 0.10, 0.25, 0.50, and 1.00. We use $\epsilon = 0.25|z|$, the smallest displacement whose damage exceeds 10% of baseline while the response still grows smoothly.

A displacement does not decay under imagination. After 40 steps it retains $6.3\times$ its initial size at $\epsilon = 0.05|z|$, $2.8\times$ at $0.25|z|$, and $1.9\times$ at $0.5|z|$, as medians over the three models.

Variance and sensitivity at different horizons. At a 40-step horizon the two rankings anticorrelate on all three models (-0.364 , -0.259 , -0.406), with damage spanning $13.5\times$ to $118.8\times$ between the most and least consequential direction (Figure 3a). The ranking replicates: split-half rank correlation of D_H has median 0.95 across the settings we tried, with a minimum of 0.52 at the smallest displacement and longest horizon. The disagreement with variance does not extend to long horizons. Pooling three models and three displacements, median $\rho(V, D_H)$ is -0.252 at horizon 20, negative in 9 of 9 settings; -0.196 at horizon 50, negative in 5 of 9; and -0.028 at horizon 100, negative in 5 of 9, with one setting reaching $+0.727$. Once a rollout has diverged, high-amplitude directions dominate the error and the rankings realign.

The correction pushes along higher-damage directions at $+0.385$, $+0.692$, and $+0.287$ (Figure 3b), without having been given that ranking. The correlations stay well below 1, so the correction is only partly concentrated there. All of this is measured on the conservative models.

Basis-independent properties reproduce better than coordinate selectors. The three models agree closely on quantities that need no choice of basis: energy recovery varies by 0.008, intervention slopes by 0.016, and the top-3 eigenvalue mass of $M_C = \mathbb{E}[gg^\top / \|g\|^2]$ with $g = \nabla C$ by 0.046 (0.448, 0.441, 0.487). Selectors on individual coordinates are less stable: ranking coordinates by ∇C and keeping the top three gives intervention slopes of -0.075 , $+0.016$, and -0.089 , beating a variance ranking on two models and changing sign across seeds. An orthogonal projector onto the top eigenspace of M_C beats the variance selector on all three models, though on one it sits at the 41st percentile of 100 random rank-3 subspaces. Since the correction is rank one at each state, the spectrum of M_C measures how much ∇C rotates as the state changes, and the amount is similar across models. This is consistent with the models encoding the same scalar in different latent coordinate systems, although we do not directly align their representations.

D Additional sweeps

Energy outside the highest-variance directions. On two of three models, principal directions 7–12 carry about twice the energy information of the top six ($2.23\times$ and $2.13\times$); on the third

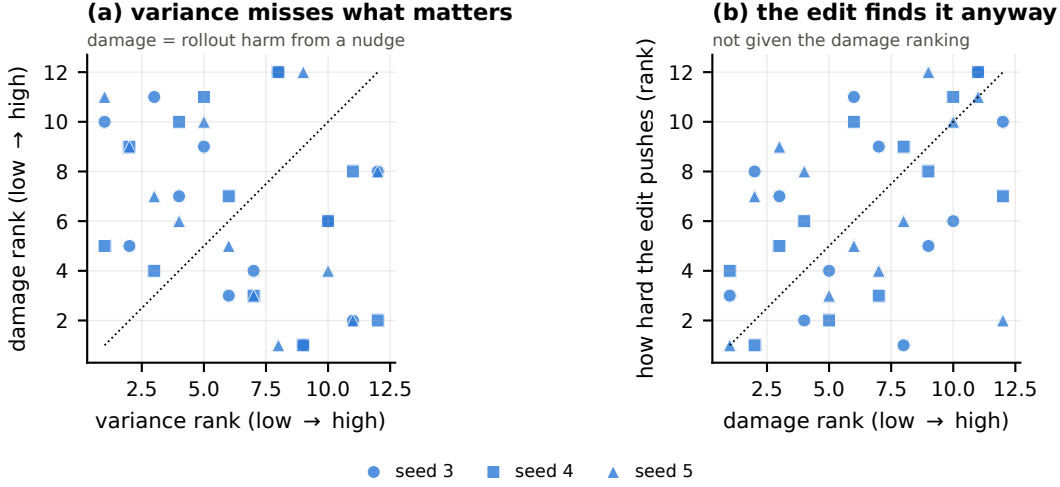


Figure 3: Each point is one of the 12 extracted directions, ranked within its own model; the dotted line marks equal ranks. (a) Variance rank against damage rank at a 40-step horizon. (b) Damage rank against the magnitude of the invariant correction along that direction.

the ratio reverses to 0.91. We report the models separately because a median would hide the sign change. Figure 5 shows the decomposition.

Accumulated invariant error. Accumulated invariant error predicts which rollouts degrade most: an integral predictor reaches $R^2 = 0.187$ against 0.021 for a fitted power law in time at the horizon where both perform best, and shuffling destroys the relation. The current experiments do not distinguish weighted from unweighted accumulation. In a nested comparison with 400 bootstrap resamples, the weighted term adds explanatory power beyond the unweighted term in 1 of 9 seed-horizon combinations, and the unweighted beyond the weighted in 0 of 9, at $n = 512$.

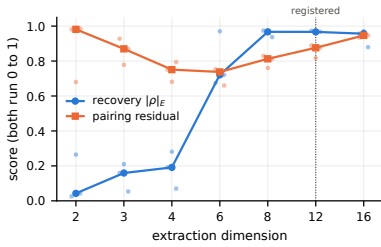


Figure 4: Energy recovery and pairing residual against extraction dimension. Lines are medians over the three models; points are individual models. Both quantities are dimensionless.

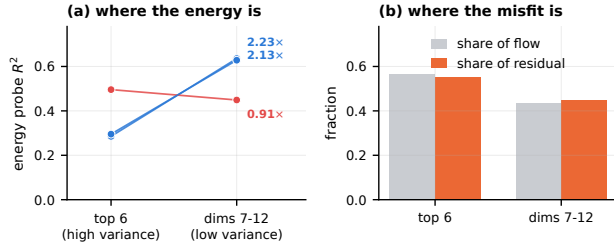


Figure 5: (a) Energy-probe R^2 for the top six principal directions against directions 7–12, per model. (b) Fraction of latent flow and pairing residual carried by each group, medians over three models.

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